THE PULLBACK OF A MODULE GENERATED BY SECTIONS IS GENERATED BY SECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-modules/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-modules/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The compatibility of the pullback with restriction to an open subspace, as the pullback along the induced morphism on the preimage. Invoked inside the counted proof at source lines 974-986 as a cross-chapter pointer. Reduces the statement to the case of a globally generated module by looking at the open pieces of a generating covering.
- The compatibility of the pullback with colimits and its right exactness. Invoked inside the counted proof at source lines 989-990 as a pointer to a lemma of this chapter. Transports the surjection exhibiting the global generators to the pullback.

Source reference. Original reference sheaves-lemma-push-pull-composition-modules: sources/stacks-modules/pinned-original-source.tex, source lines 985–985. Navigation pointer only: the counted text refers to the Stacks item sheaves-lemma-push-pull-composition-modules, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-exactness-pushforward-pullback: sources/stacks-modules/pinned-original-source.tex, source lines 990–990. Navigation pointer only: the counted text refers to the Stacks item lemma-exactness-pushforward-pullback, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. The pullback $f^*\mathcal{G}$ is locally generated by sections if $\mathcal{G}$ is locally generated by sections.*

Proof. Given an open subspace V of Y we may consider the commutative diagram of ringed spaces

$$\begin{array}{ccc} (f^{-1}V, \mathcal{O}_{f^{-1}V}) & \xrightarrow{j'} & (X, \mathcal{O}_X) \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ (V, \mathcal{O}_V) & \xrightarrow{j} & (Y, \mathcal{O}_Y) \end{array}$$

We know that $f^*\mathcal{G}|_{f^{-1}V} \cong (f')^*(\mathcal{G}|_V)$, see Sheaves, Lemma sheaves-lemma-push-pull-composition-modules. Thus we may assume that $\mathcal{G}$ is globally generated.

We have seen that f^* commutes with all colimits, and is right exact, see Lemma lemma-exactness-pushforward-pullback. Thus if we have a surjection

$$\bigoplus\nolimits_{i \in I} \mathcal{O}_Y \to \mathcal{G} \to 0$$

then upon applying f^* we obtain the surjection

$$\bigoplus\nolimits_{i \in I} \mathcal{O}_X \to f^*\mathcal{G} \to 0.$$

This implies the lemma. □